

Network Security Starter Plan

Expanded Edition – From Zero to Job-Ready Foundations

Cybersecurity & Carrier / Telecom Security

June 2026 | Windows 10/11 | 100% Free Tools

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1. 90-Day Roadmap

Week	Focus	Daily tasks (2 hrs/day)	Deliverable
1-2	Network Basics	Day 1-3: Cisco Intro to Cybersecurity. Day 4-7: IP, subnet masks, Hubs, gateway, DNS. Day 8-14: OSI model, ping/traceroute	Homework
3-4	TCP/IP	DNS, HTTP/HTTPS, ports, TCP handshake. Practical Networking playlist Wireshark Lab 2	Wireshark Lab 2
5-6	Security Fundamentals	Cyber basics, AES/TLS, firewalls, WPA3, VPNs. TryHackMe Network Security Essentials	Network Security Essentials
7-8	Windows & Logs	Event Viewer, PowerShell Get-Net*, syslog basics, NetworkChuck's Analysis Fundamentals	NetworkChuck's Analysis Fundamentals
9-10	Defense Tools	Wireshark filters, Nmap scanning, Cisco Packet Tracer – build a router config	Router config
11	SOC / Carrier	SOC 101 – SIEM, alert triage, incident playbooks, BGP/OSPF in a SOC	SOC notes
12	Certify	ISC2 CC practice exam, build 1-page resume with 3 labs, LinkedIn profile	CC profile

2. Free Learning Platforms

Platform	Course	URL
Cisco Networking Academy	Intro to Cybersecurity, Network Technician Career Path	netacad.com / skillsforall.com
TryHackMe	Network Security Essentials (free)	tryhackme.com
ISC2	Certified in Cybersecurity – free exam	isc2.org
Fortinet Training	NSE 1-3	training.fortinet.com
IBM SkillsBuild	Cybersecurity Fundamentals	skillsbuild.org
Microsoft Learn	SC-900 Security Fundamentals	learn.microsoft.com
NCERT / DIKSHA India	Device & Network Security Nano course	diksha.gov.in

3. YouTube Study Order

Weeks 1-4 – Fundamentals: Practical Networking – Networking Fundamentals | David Bombal | NetworkChuck | Professor Messer – Network+ | Computerphile

Weeks 5-12 – Security Labs: The Cyber Mentor – SOC 101 | John Hammond | LiveOverflow | ippsec | Hak5 | SANS Institute | Null Byte | Cybrary

4. Windows Lab Environment Setup

1. Wireshark – wireshark.org – install with Npcap
2. Nmap – nmap.org – default install
3. Cisco Packet Tracer – netacad.com – free with Cisco account

All run fine on a normal laptop, 8GB RAM is enough. Create a folder C:\NetSec-Labs for all captures.

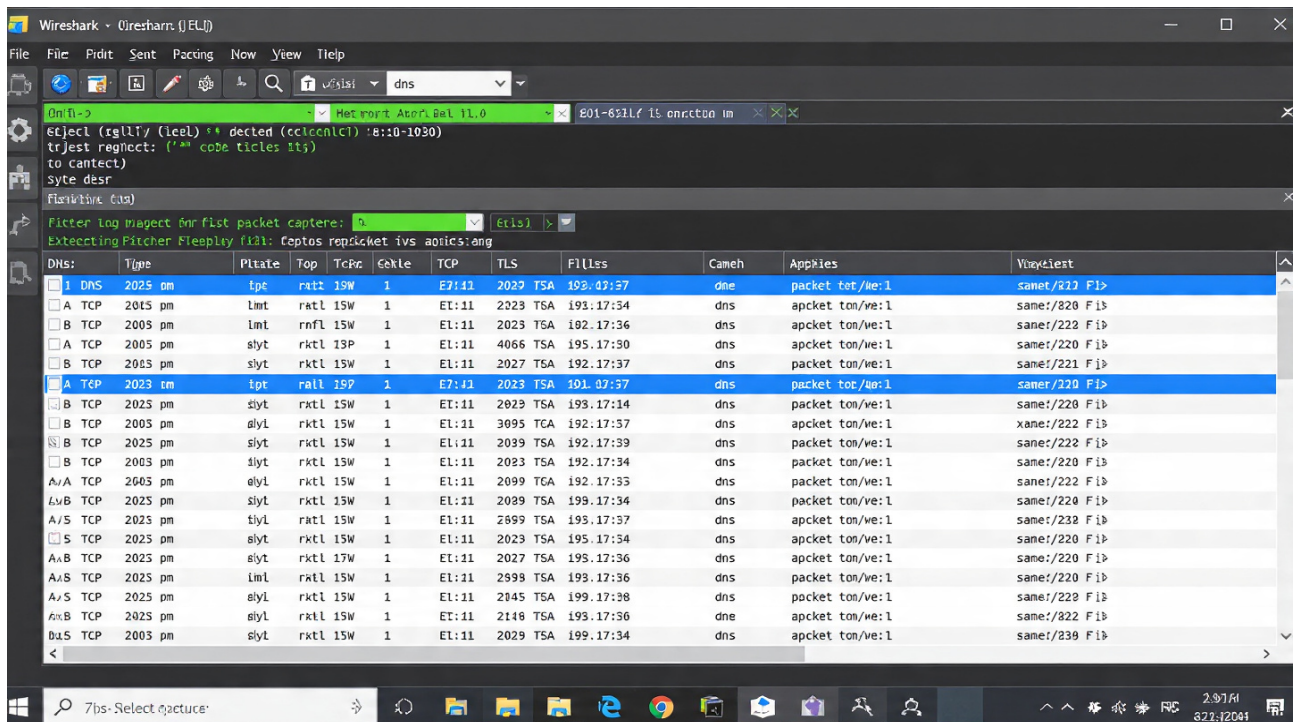
5. Lab 1 – Network Mapping

Goal: See your own ISP path

1. Win+R → cmd
2. ipconfig /all – note IPv4, Gateway, DNS
3. ping 8.8.8.8 -n 4
4. ping google.com -n 4
5. tracert google.com
6. arp -a

Write down: your local IP, router IP, first 3 ISP hops from traceroute, DNS server. This is your network baseline.

6. Lab 2 – Wireshark Packet Capture – Step by Step



- Step 1:** Open Wireshark → select Wi-Fi / Ethernet → Start
- Step 2:** In browser open <http://neverssl.com> then <https://example.com>
- Step 3:** Stop capture. Filter: dns – see your DNS queries
- Step 4:** Filter: tcp.port==443 – right-click a packet → Follow → TLS Stream
- Step 5:** Filter: tcp.flags.syn==1 – this is the TCP 3-way handshake
- Save as lab2-dns-tls.pcapng in C:\NetSec-Labs

7. Lab 3 – Nmap Safe Scanning

```

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Only scan your own home network. Never scan public IPs.

```

nmap -sn 192.168.1.0/24 # find live hosts
nmap -sV -O 192.168.1.1 # router version detection
nmap -p 22,80,443,3389 192.168.1.0/24

```

Document: IP, open ports, service versions. Close any unnecessary ports on your router admin page.

8. Lab 4 – Windows Firewall Test

Windows Security → Firewall & network protection → Advanced settings → Outbound Rules → New Rule → Port → TCP 80 → Block.

Try browsing <http://neverssl.com> – it should fail. Check Monitoring → Firewall log. Then Disable the rule. You just tested a perimeter control like carriers do.

9. Lab 5 – DNS Security Comparison

```
nslookup google.com
```

```
nslookup google.com 1.1.1.1
```

```
nslookup google.com 9.9.9.9
```

Compare response times and IPs. 1.1.1.1 = Cloudflare, 9.9.9.9 = Quad9 with malware blocking. This is DNS security in practice.

10. OSI Model Deep Dive

L	Name	PDU	Devices	Security
7	Application	Data	Proxy	TLS, WAF
6	Presentation	Data	–	Encryption
5	Session	Data	–	Session timeout
4	Transport	Segment	Firewall	Port filtering
3	Network	Packet	Router	ACLs, IPsec
2	Data Link	Frame	Switch	MACsec, WPA3
1	Physical	Bits	Cable	Physical locks

11. TCP/IP and 3-Way Handshake

SYN → SYN-ACK → ACK. Then data flows. FIN closes gracefully. RST aborts. In Wireshark:

`tcp.flags.syn==1` and `tcp.flags.ack==0` finds new connections. This is how SOC analysts spot port scans.

12. Ports & Protocols Master Table

Port	Proto	Use	Secure?	Note
22	SSH	Remote shell	Yes	Replace Telnet
23	Telnet	Remote shell	No	Never use
25/587	SMTP	Email send	TLS only	587 submission
53	DNS	Name resolution	DoH/DoT	853 = DoT
67/68	DHCP	IP assign	Snooping	Rogue DHCP risk
80	HTTP	Web	No	Redirect to 443
110/143	POP3/IMAP	Email	995/993 TLS	Use TLS
443	HTTPS	Secure web	Yes	TLS 1.2+
3389	RDP	Remote desktop	VPN+MFA	Never public
161	SNMP	Monitoring	v3 only	v1/v2c insecure
389/636	LDAP	Directory	636 = LDAPS	Use LDAPS

13. Subnetting Worked Examples

Formula: $\text{hosts} = 2^{(32-\text{mask})} - 2$

192.168.1.0/24 → 256 addresses, 254 hosts, mask 255.255.255.0

/25 → 128 / 126 hosts

/26 → 64 / 62

/27 → 32 / 30

/28 → 16 / 14

/29 → 8 / 6

/30 → 4 / 2 – point-to-point links

Private ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16

Example: need 50 hosts → /26 gives 62, fits. Network: 192.168.1.0/26, usable .1-.62, broadcast .63

14. Wireshark Filter Cheat Sheet

Filter	What it finds
dns	All DNS
http	HTTP only
tls.handshake.type == 1	TLS ClientHello
tcp.port==443	HTTPS traffic
ip.addr==192.168.1.1	Traffic to/from router
tcp.flags.syn==1 and tcp.flags.ack==0	New TCP connections
icmp	Ping packets

15. Nmap Cheat Sheet – Home Lab Only

Command	Purpose
nmap -sn 192.168.1.0/24	Ping sweep – find live hosts
nmap -sV 192.168.1.1	Service versions
nmap -O 192.168.1.1	OS detection
nmap -p 22,80,443 192.168.1.0/24	Specific ports
nmap -sS -T4 192.168.1.1	Stealth SYN scan, fast

Warning: Only scan networks you own. Scanning others without permission is illegal in India and most countries.

16. Carrier / Telecom Security Career Path

Path: NOC Analyst → SOC Analyst → Network Security Engineer → Security Architect

Entry certs: Cisco CCNA, ISC2 Certified in Cybersecurity, CompTIA Security+

Carrier skills: BGP, OSPF, MPLS, firewalls, DDoS mitigation, RADIUS/TACACS+, NetFlow

Portfolio tip: 3 Wireshark pcaps with analysis + 1 network hardening report beats a certificate alone.

Where to apply: ISPs, telecom NOCs, MSSPs, banks, cloud SOC's, government CERT teams.

17. Glossary

Term	Meaning
OSI	7-layer model for how networks work
TCP/IP	Core internet protocol suite
DNS	Domain Name System – translates names to IPs

Term	Meaning
TLS	Transport Layer Security – encrypts web traffic
VPN	Virtual Private Network – encrypted tunnel
Firewall	Filters traffic by port/IP/rules
SIEM	Security Information Event Management – SOC tool
NOC	Network Operations Center – carrier monitoring
SOC	Security Operations Center – threat monitoring
PCAP	Packet capture file from Wireshark

18. 12-Week Study Tracker

Week	Hours planned	Labs done	Notes / What I learned
1	_____	_____	
2	_____	_____	
3	_____	_____	
4	_____	_____	
5	_____	_____	
6	_____	_____	
7	_____	_____	
8	_____	_____	
9	_____	_____	
10	_____	_____	
11	_____	_____	
12	_____	_____	

Network security rewards patience and hands-on reps. Do the 5 labs, save your pcaps, finish Cisco Intro + ISC2 CC. That is your first resume line. Good luck!

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